XN Project 2 - Finding Novel Intrinsic Oncology Targets and Biology

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December 12, 2023

Abstract

In this short project, we create comprehensive computational analyses that connect specific groups of genes to various types of cancers. We identify cancer vulnerabilities and classify predictive models that share common genetic determinants using CRISPR KO gene dependencies, mutational and copy number profiles and RNA expression data obtained from over 1000 cancer cell lines. Multivariate feature importance showed that a combination of mutation, gene expression and network embedding features are informative for target prediction in most cancer types. This paper does not contain new research ideas but it aims to provide information to the industry researchers on the relation between different genes or groups of genes and the risk of cancer.

1 Introduction

Merck & Co., Inc., an American multinational pharmaceutical company headquartered in New Jersey, has been working in conducting analyses of the genetic and pharmacological features of an extensive range of human cancer models. The company has been interested in mining publicly available genomic data sets 1. to identify cancer vulnerabilities, differential cancer gene dependencies, 2. to classify predictive models that share common genetic determinants using CRISPR KO gene dependencies, mutational and copy number profiles and RNA expression data obtained from over 1000 cancer cell lines.

Aiming towards this objective, we create comprehensive computational analyses that connect specific pharmacological weaknesses to genomic patterns and translate integrated genomics from cell lines into the stratification of cancer patients.

1.1 Our Objective

In this project, we use publicly available data set focused on genes of various patients and apply machine learning algorithms to better understand how certain genes or groups of genes relate to various types of cancers. We work with the information and advice provided by the industry advocate working at Merck Research Laboratories and aim to understand the data as well as how the results of this project would be used in the industry to help real cancer cases. The main application of this project is to provide more information to industry researchers on the relation between different genes or groups of genes and the risk of cancer. This is an important task as finding ways to develop medicines that effectively eliminate cancer cells would save millions of lives.

1.2 Terminology

Before we describe the data sets, we will need some terminology:

- **CCLE (Cancer Cell Line Encyclopedia):** is a multi-institutional collaboration project, dated from 2006, aiming to link genes to cancer types but for large-scale genetic characterization of 1000 cancer cell lines. After expansion of each cell, the gene effect is collected in generated data, and the goal is to relate these genes to varioups cancer types.
- **KO (knockout):** is a technology that research scientists use to selectively modify the DNA of living organisms; i.e., the DNA is mutated in a way that stops the gene's expression permanently.
- **CRISPR (clustered regularly interspaced short palindromic repeats):** is a technology that research scientists use to selectively modify the DNA of living organisms, and it is the fastest and most direct approach of gene knockout.

1.3 Data sources

The data files used in this project were downloaded from the Depmap Portal website. Despite the recent data (2023) available, we used the Year-2022-Q2 files due to lack of explained methodologies for the recent data (2023). We focus on two data sets, namely: **CRISPR-gene-dependency** and **CCLE-expression**. Then for each data type we use various machine learning algorithms and two classifiers, namely: **lineages** and **primary diseases** that are depended on essential genes for tumorigenesis.

2 Analysis

Literate review suggests that computational intelligence methods can be applied to predict novel therapeutic targets in oncology. Very simple models, incorporating only gene expression, mutation and essentially information for individual genes, already achieved reasonable prediction performance for some cancer types. Multivariate feature importance showed that a combination of mutation, gene expression and network embedding features are informative for target prediction in most cancer types. Therefore, we used some of the algorithms learned in our Machine Learning 1 class to analyze the data. The models used can be found in the preceding section.

2.1 Methodology

- **Logistic Regression** is one of the most basic classification algorithm that is know to provide good result despite its simplicity. We used this algorithm because it is easy to implement and has the potential to be pretty accurate. Logistic regression is a statistical method used for binary classification, predicting the probability that an instance belongs to a certain category. It works by fitting a logistic curve (sigmoid function) to the data.

- **Linear Discriminant Analysis (LDA)** is another classifier we used to find a linear combination of genotypes which separates the multiple classes of cancer types. The model creates multiple linear discrimication function to distinguish amoung the data. The aim of LDA is to maximize the ratio of the between-group variance and the within-group variance. Also, it is worth mentioning that because LDA is based on classic estimators of location and covariance, it is sensitive to outlying samples (which decrease the LDA performance with an increase in the number of incorrectly assigned samples).

- **Clustering** is an unsupervised machine learning method that groups unlabelled data into categories, or clusters. This method is evaluated by how similar within-cluster data is and how different between-cluster data is. In this project, we use the hierarchical clustering and the density based clustering models. The performance metric we used is the Calinski-Harabasz index (CH) which is given by the formula below, in which, $\mathbf{x_i}(1 \leq i \leq n)$ is our data, $\mathbf{C_i}(1 \leq i \leq k)$ are the clusters, and $\mathbf{c_i}$ are their respective centroids. $\mathbf{c}$ is the overall centroid.

$$BCSS = \sum_{i=1}^{k} n_i ||\mathbf{c}_i - \mathbf{c}||^2 \qquad WCSS = \sum_{i=1}^{k} \sum_{\mathbf{x} \in C_i} ||\mathbf{x} - \mathbf{c}_i||^2 \qquad CH = \frac{BCSS/(k-1)}{WCSS/(n-k)}$$

- **Random Forest Classification** Random Forest Classifiers seemed very useful for the problem as using the results from a multitude of decisions trees could be very effective as the data set is large and overall leads to more accurate results than a simpler decision tree approach. The application of Random Forest Classifiers used in this problem was using Sklearns Random Forest Classifier in conjunction with the OneVsRest Classifier. Random Forest Classifier are known to be effective for large numbers of classes. The OneVsRest Classifier is recommended by the library for when the data set features a large number of classes with limited samples. This model fits one classifier per class than fits it against all other classes hence the one class vs the rest of the classes. The data was also modified using the MutualInfoClassfier. This feature selection tool from Sklearn uses the independence and dependence between features to perform feature selection on the data. This classifier was used to select the 600 most independent features to be used in the classification. Using these tools the RandomForestClassifier was applied to the data set, see results below.

- **Neural Nets** Neural Nets are well known in Machine Learning fields as a very flexible tool that can almost always be applied. The flexibility of neural networks is offset by the difficulty to logically understand the model. Many times neural networks are viewed as "black box" where the user will not know what is really going on under the hood. For this problem a simple model similar to ones presented in class' discussions was used. TensorFlow and Keras were used to build the Neural Network. The network featured 4 Dense layers which lead to a total of 9764380 trainable parameters. The data set did struggle with over-fitting due to the lack of samples for certain classes as well as limited size of the data set. This lead to the implementation of 3 dropout layers with parameter of .3, .25, and .25 respectively. Smaller batch sizes were also implemented in order to ensure there was proper noise present in the training. Both of these lead to a model with significantly less over fitting. The overall architecture of the model was 7 layers, 4 dense and 3 dropouts layers. This model was used between all datasets and performed remarkably well see results below.

2.2 Results

- **Logistic Regression** We fit our logistic regression model to four different combinations of dataframes; CRISPR(primary disease classifier), CRISPR(lineage classifier), CCLE(primary disease classifier), CCLE(lineage classifier). In addition, we tested the following number of max iterations to pass into the logistic model: [50,100,400,1000]. The logistic regression scores for a maximum of 400 iterations were as follows: 0.663 for CRISPR Primary Disease, 0.663 for CRISPR Lineage, 0.778 for CCLE Primary Disease, and 0.770 for CCLE Lineage. The best fit result for logistic regression was using the CCLE Primary Disease data with maximum iterations equal to 400, and the accuracy score was 0.778. A summary table can be found in the next page.

Iterations	CRISPR : Primary Disease	CRISPR : Lineage	CCLE: Primary Disease	CCLE: Lineage
50	0.620	0.620	0.707	0.705
100	0.644	0.644	0.753	0.753
400	0.663	0.663	0.778	0.770
1000	0.663	0.663	0.778	0.770

- **Linear Discriminant Analysis (LDA)** After testing the four datasets, we got the following LDA scores: 0.517 for CRISPR Primary Disease, 0.509 for CRISPR Lineage, 0.736 for CCLE Primary Disease, and 0.739 for CCLE Lineage.

CRISPR : Primary Disease	CRISPR : Lineage	CCLE: Primary Disease	CCLE: Lineage
0.517	0.509	0.736	0.739

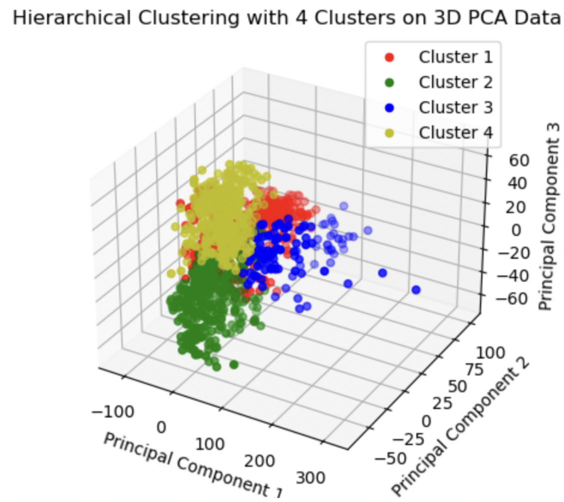
  The LDA results were overall consistent with the logistic regression but weaker than the logistic regression. This could be due to the fact that LDA assumes that all classes are linearly separable while the logistic regression does not assume any specific shapes of densities in the space of predictor variables. Also LDA being sensitive to outliers translates to weaker scores than the logistic model. Logistic regression makes significantly weaker assumptions. So, it is more robust and less sensitive to incorrect modeling assumptions that other models might require.

- **Clustering** After testing on the CCLE Expression and the CRISPR Dependency datasets, we observe that Principal Component Analysis (PCA) vastly improved the performance of the hierarchical clustering model, the CH scores of which are shown in the first table. The performance of the density based clustering model is worse in comparison. Without PCA, this model failed to segment the dataset into meaningful clusters, in other words, almost every point is its own cluster. Even after performing PCA, its CH scores are on the lower side, as seen on the second table.

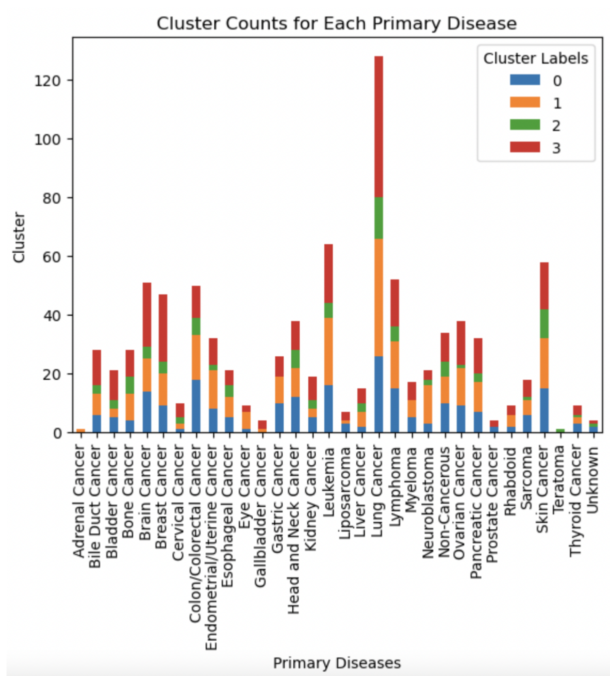
CCLEexpr before PCA	CCLEexpr after PCA	CRISPRdepend before PCA	CRISPRdepend after PCA
14.3049	101.45	4.1767	60.999

CCLEexpr	CRISPRdepend
24.366	10.195

Below is the output of the hierarchical clustering model on the training dataset.



We cannot obtain many insights from the clustering models. As it can be seen in the bar graph below, the clusters failed to segment primary diseases into meaningful groups, i.e., almost every cluster contains almost every primary disease.



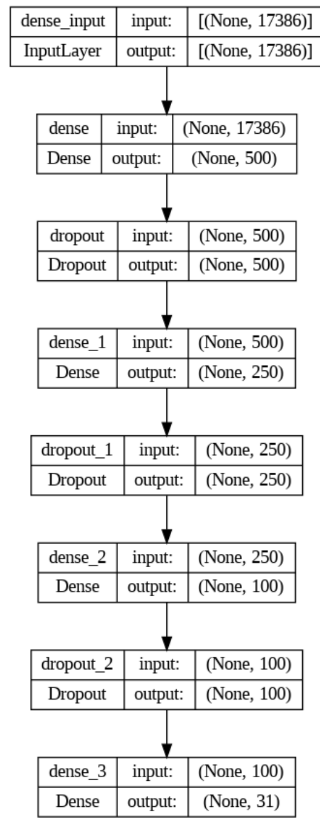
- **Random Forest Classification** Tested the 4 datasets, results as follows CCLE: Primary Disease Acc:0.4875 F1:0.6025, CCLE Lineage: Acc:0.5108 F1:0.6275, CRISPR Primary Disease: Acc:0.3563 F1:0.4513 CRISPR Lineage: Acc:0.3619 F1:0.4641. These results are as expected and are in line with the other models performance. The two metrics used were accuracy and F1 score. Accuracy is just the typical accuracy metric of correct predictions over all predictions. F1 score is the harmonic mean of recall and precision. The formula for F1 score is given as $F1 = (2 * Recall * Precision)/(Precision + Recall)$. The F1 score is the better indicator of the performance of the RandomForestClassifier and it is in line with the results of the other models tested. This shows RandomForestClassifiers are a valid model for the classification of cancer types using all 4 of the data sets. Though RandomForest is an applicable model it yielded the worst results indicating a different model would likely be a better approach.

	CRISPR: Primary Disease	CRISPR: Lineage	CCLE: Primary Disease	CCLE: Lineage
Accuracy	0.3563	0.3619	0.4875	0.5108
F1 Score	0.4513	0.4641	0.6025	0.6275

- **Neural Nets** Tested the 4 datasets, results as follows CCLE: Primary Disease Train Acc:0.8069 Test Acc:0.7234, CCLE Lineage: Train Acc:0.8532 Test Acc:0.7340, CRISPR Primary Disease: Train Acc:0.6262 Test Acc:0.5438 CRISPR Lineage: Train Acc:0.7477 Test Acc:0.5576.

	CRISPR: Primary Disease	CRISPR: Lineage	CCLE: Primary Disease	CCLE: Lineage
Training	0.6262	0.7477	0.8069	0.8532
Validation	0.5438	0.5576	0.7234	0.7340

These results are in line with the other models. There were some issues with over fitting when training the neural net. This is likely caused by the combination of limited number of cells to train on from the original data sets as well as the lack of samples for some of the given classes. The architecture of the neural net is shown below.



This same neural net was applied to all 4 data sets to allow for direct comparisons between results. As shown in the above figure the model featured 3 dropout layers to help restrict the over fitting of the training data. The dropout layers had parameters of .3, .25, .25 respectively. A smaller batch size was also used when training to generate more noise and reduce over fitting. After using these normalization techniques the validation accuracy was much more comparable to the training accuracy showed the effectiveness of the normalization techniques. The best performing predictor of cancer type was CCLE lineage which is in line with the other models. Overall the neural net had great results and could be a very effective model when applied to this problem set.

3 Conclusion

After testing a multitude of models we found that each model works as a good predictor of the categorical data. Looking at the fact that Lineage has 29 classes and the primary disease data 33, a random guessing would expect an accuracy of 1/30. As each and every model we tested performs significantly better than this, it shows that the models are effective at guessing the correct categories.

An issue that could lead to lowing accuracy of the classification of the tumorous is the similarities between cancer types. Also many cancers may have similar gene expressions so the model may guess wrong on these similar cases, but that wrong guess may not be that far off from the real result depending on cancer similarity. Comparing the performance scores of the models we run, the logistic regression model performs better than all other models.

Despite the above mentioned issues and similarities that some cancer types have, the logistic regression makes significantly weaker assumptions. So, it is more robust and less sensitive to incorrect modeling assumptions that other models might require. Also, we observe that the CCLE data set has better testing results than the CRISPR one; this could be due to the fact that CCLE has more data points and also more features (genotypes) than the CRISPR data set. Even though the models run predict similar results, an increased accuracy of 1% to 2% in cancer classifying translates to certain treatment regimes, and therefore could save thousands of lives.

References

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DepMap data downloads. (n.d.). https://depmap.org/portal/download/all/